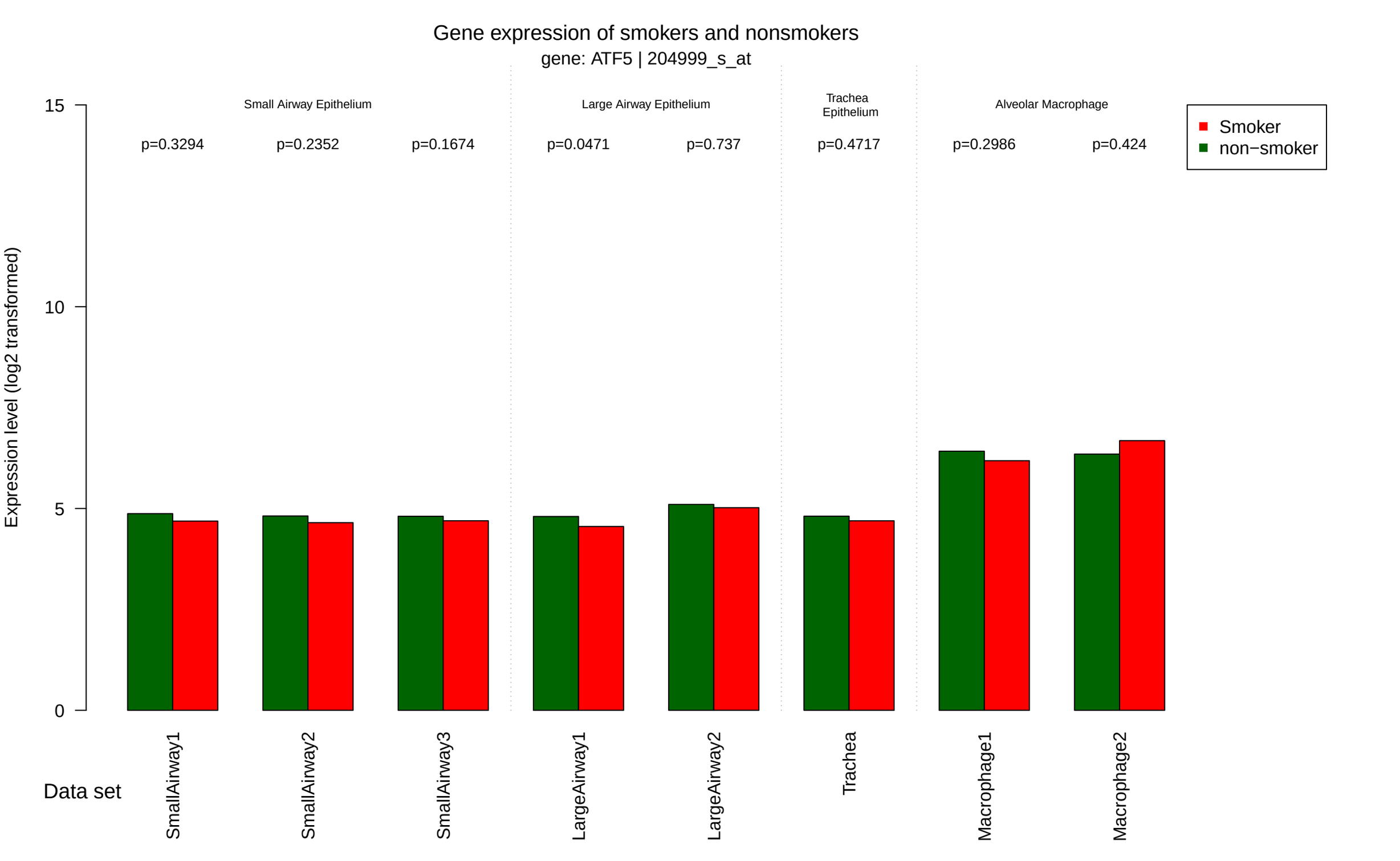
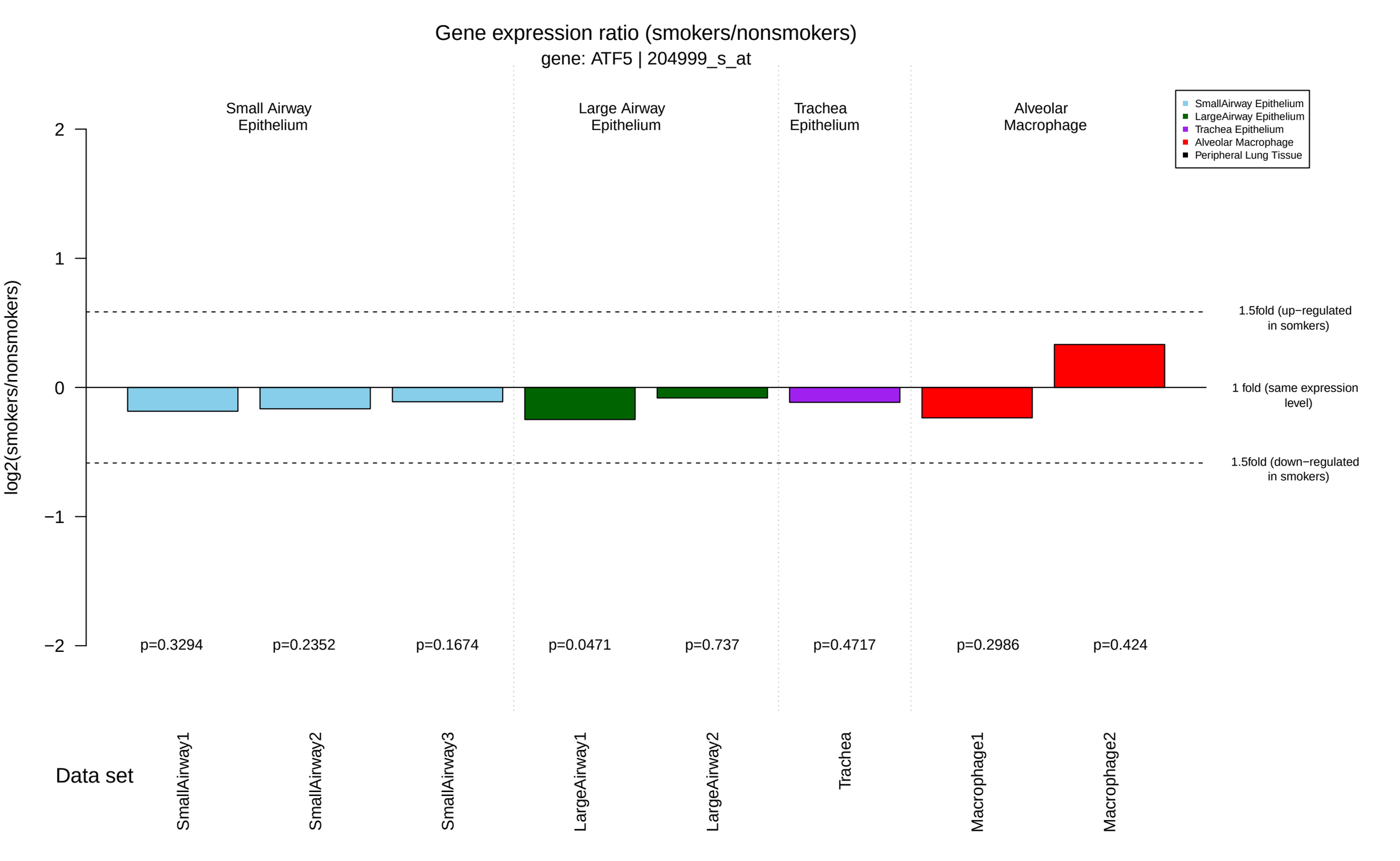
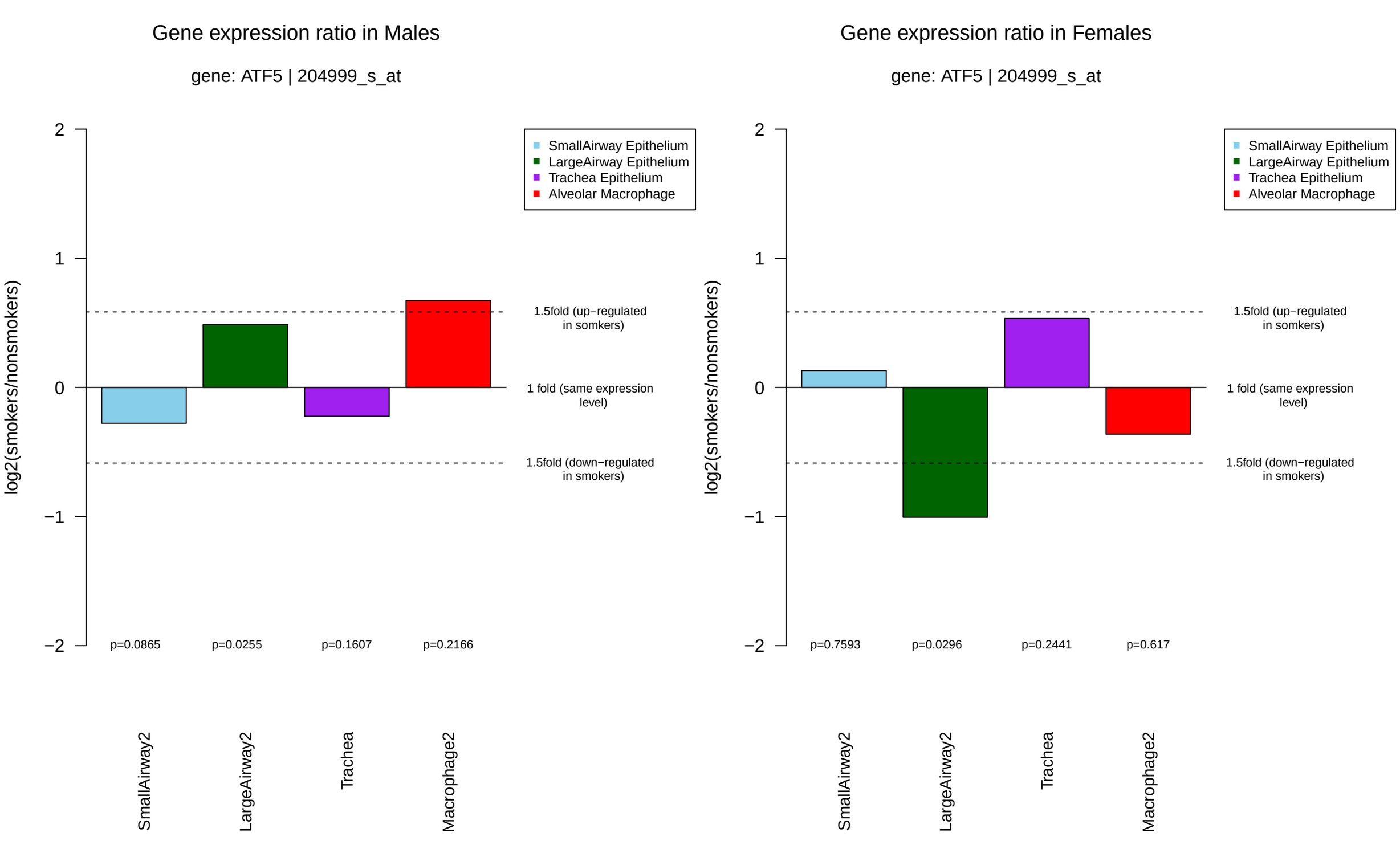
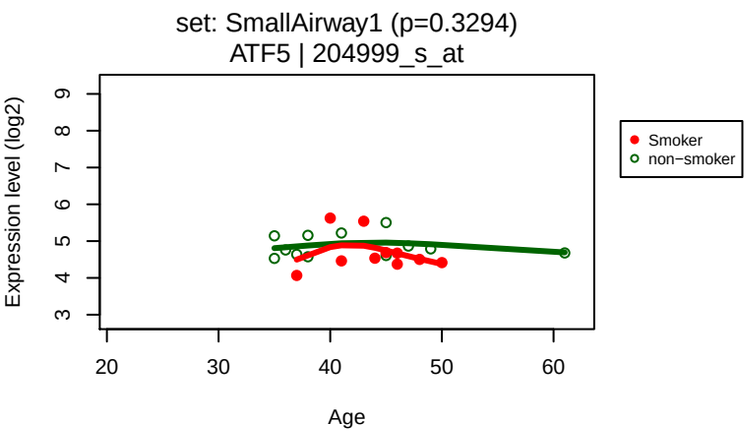
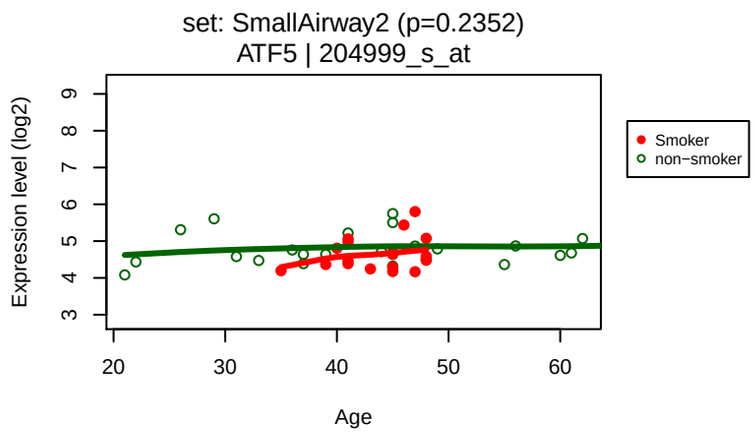
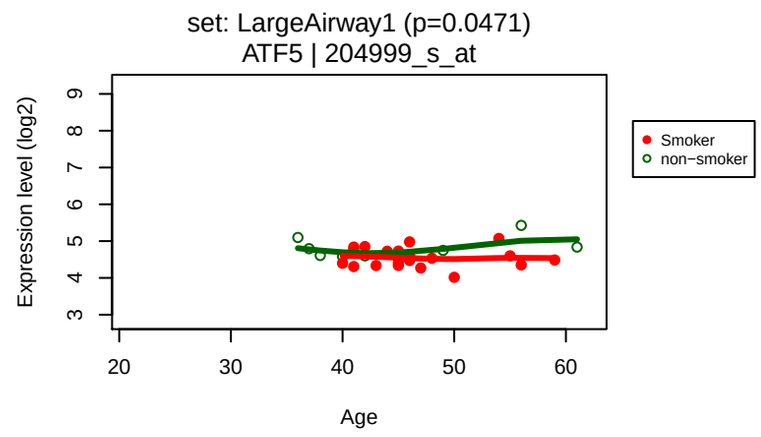

Gene expression of smokers and nonsmokers
gene: ATF5 | 204999_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3294
p=0.2352
p=0.1674
p=0.0471
p=0.737
p=0.4717
p=0.2986
p=0.424
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

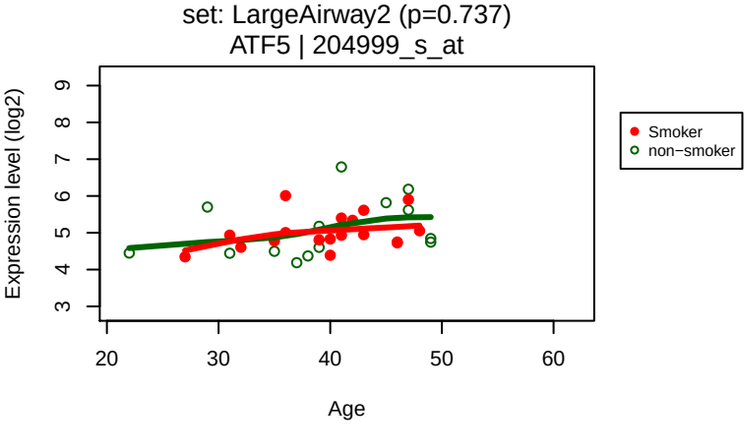

Gene expression ratio (smokers/nonsmokers)
gene: ATF5 | 204999_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3294
p=0.2352
p=0.1674
p=0.0471
p=0.737
p=0.4717
p=0.2986
p=0.424
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

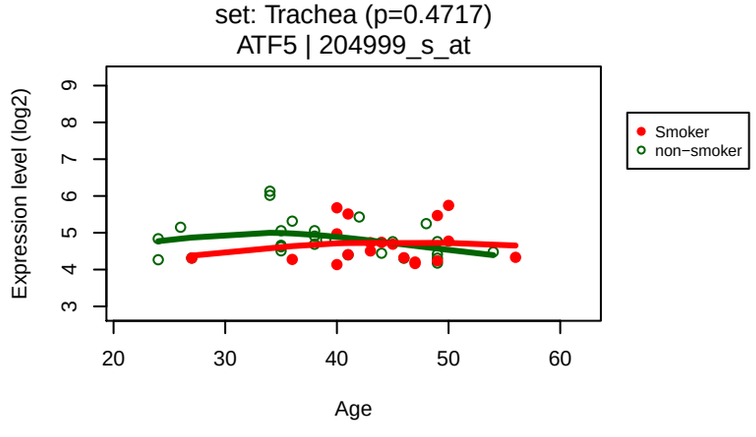

Gene expression ratio in Males
gene: ATF5 | 204999_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0865
p=0.0255
p=0.1607
p=0.2166
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF5 | 204999_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7593
p=0.0296
p=0.2441
p=0.617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

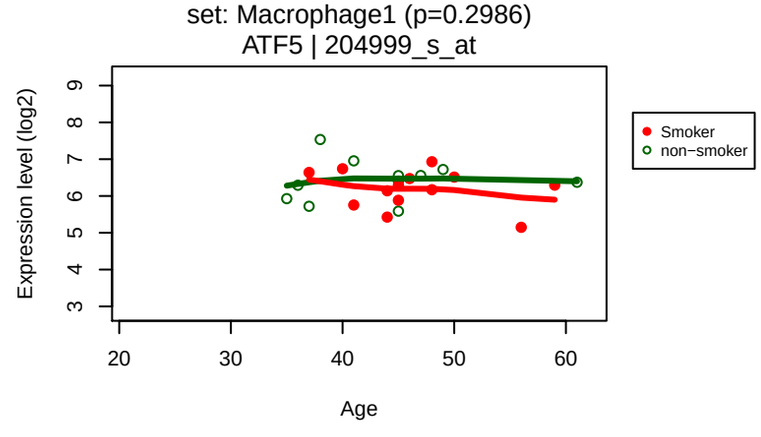

set: SmallAirway1 (p=0.3294)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker

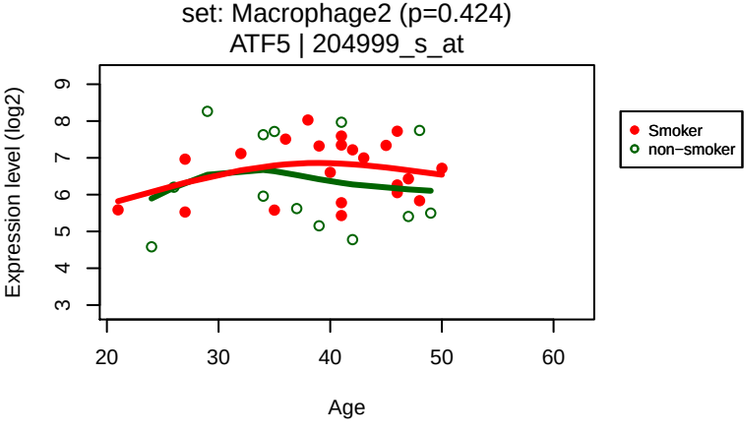

set: SmallAirway2 (p=0.2352)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker

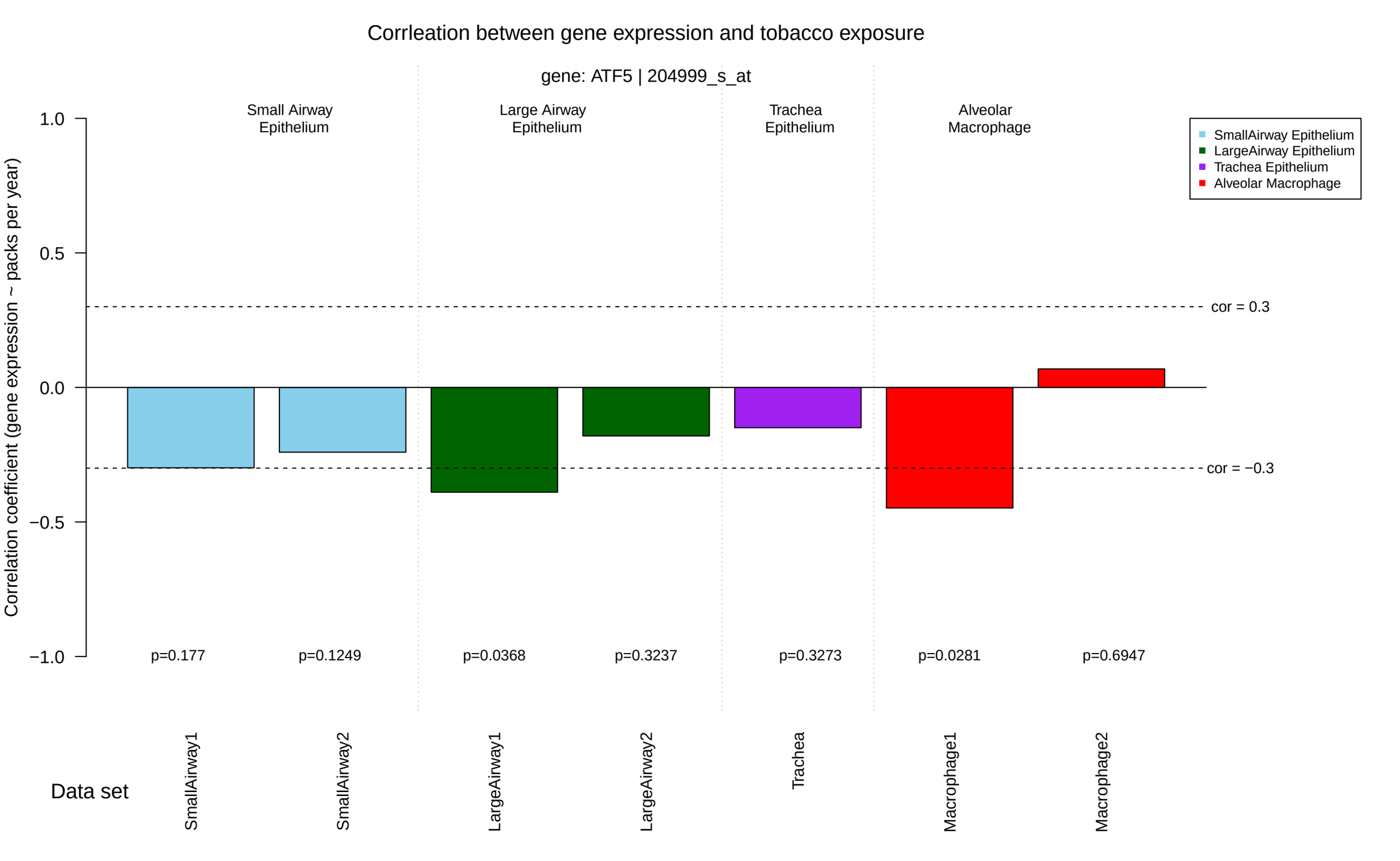

set: LargeAirway1 (p=0.0471)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.737)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4717)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2986)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.424)
ATF5 | 204999_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF5 | 204999_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.177
p=0.1249
p=0.0368
p=0.3237
p=0.3273
p=0.0281
p=0.6947
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2